

Buy it Now, Track Me Later: Attacking User Privacy via Wi-Fi AP Online Auctions

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Online Tracking and Privacy

PURPOSE OF THE STUDY

- Do product photos leak MAC addresses?
- Can those MAC addresses be located?
- Can the devices be tracked between owners?



METHODS

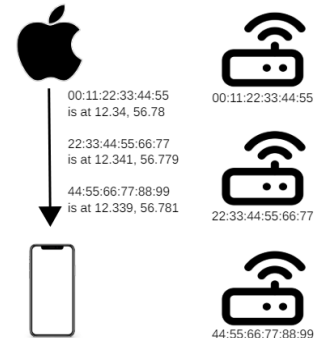
- 788,000 eBay listings using eBay API
- Detected and OCR'd label text
- Extracted and verified MACs
- Queried Apple's WPS for coordinates
- Compared locations over time



Extract MAC
address



Query
Apple's WPS



) The Apple WPS responds with the BSSID's geolocation as well as the geolocations of up to 400 additional nearby BSSIDs.

Retrieve
locations



MAIN RESULTS

- 159,000 MACs extracted
- 13,000 geolocated
- 5,628 seller locations
- 3,123 buyer locations
- 439 device movements detected

Table 2: Top Wi-Fi AP manufacturers derived from eBay AP photographs containing BSSIDs.

# MAC Addresses	%	Manufacturer
31,820	20.0	Netgear
17,768	11.1	Cisco
11,451	7.2	Belkin
11,132	7.0	TP-Link
8,069	5.1	Ubiquiti
79,133	49.7	2,536 other
159,373	100	Total

Table 3: The number of eBay auctions per seller country in our corpus.

# Auctions	%	Country
105,489	73.1	US
20,228	14.0	GB
4,684	3.2	DE
3,426	2.4	CN
2,666	1.9	AU
7,864	5.5	142 other countries
144,357	100	Total

WHY IS IT A RISK?

- No hacking needed
- Works at massive scale
- Photos leak more than expected

RECOMMENDATIONS

- Blur labels
- Marketplaces should detect identifiers
- Manufacturers should not print MACs



CRITICAL REVIEW

SHORTCOMINGS



WPS Implementation



eBay API



Computer Vision Pipeline



Case Studies



Ethical Discussion



Relevance



MAC Address Validation

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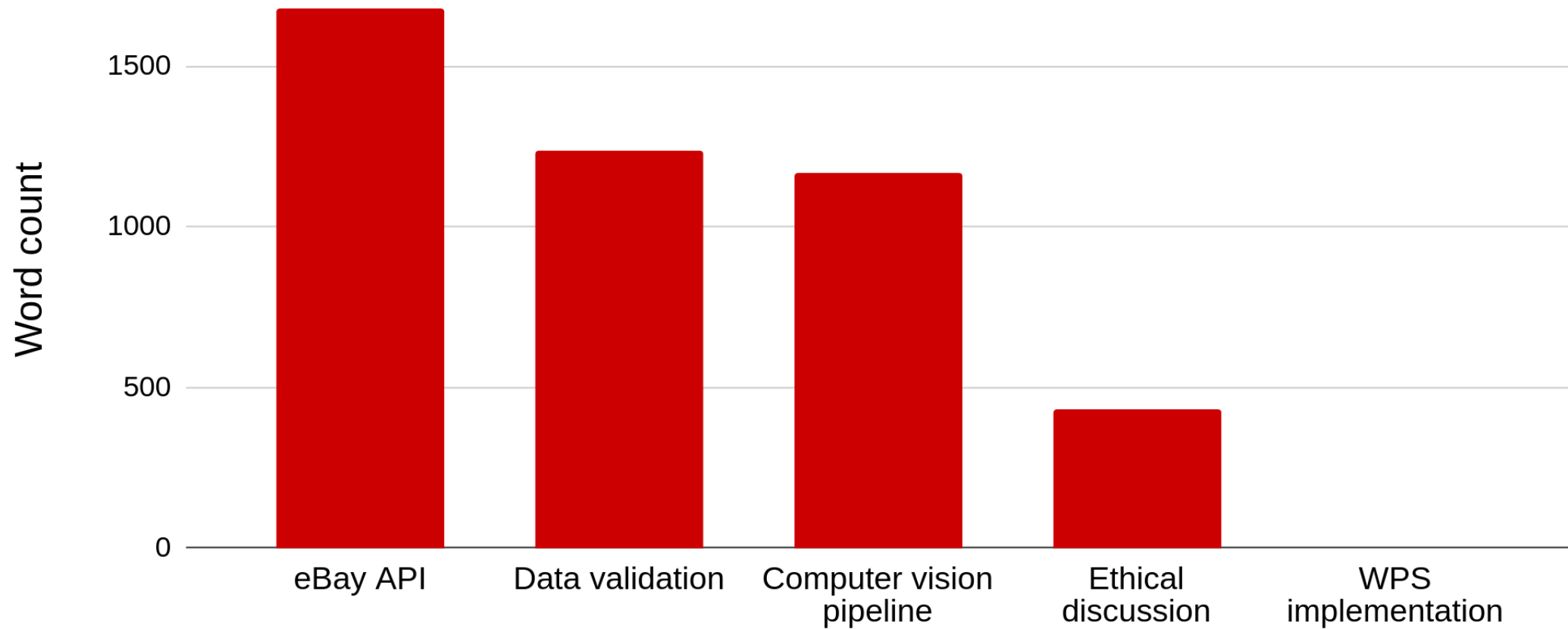


Relevance



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SHORTCOMINGS: CASE STUDIES



DEVICES FROM
CONFLICT AREAS



USED ITEMS SOLD AS
NEW



GRANULAR MODEL
INFERENCES



AUCTIONED DEVICES IN
SENSITIVE LOCATIONS

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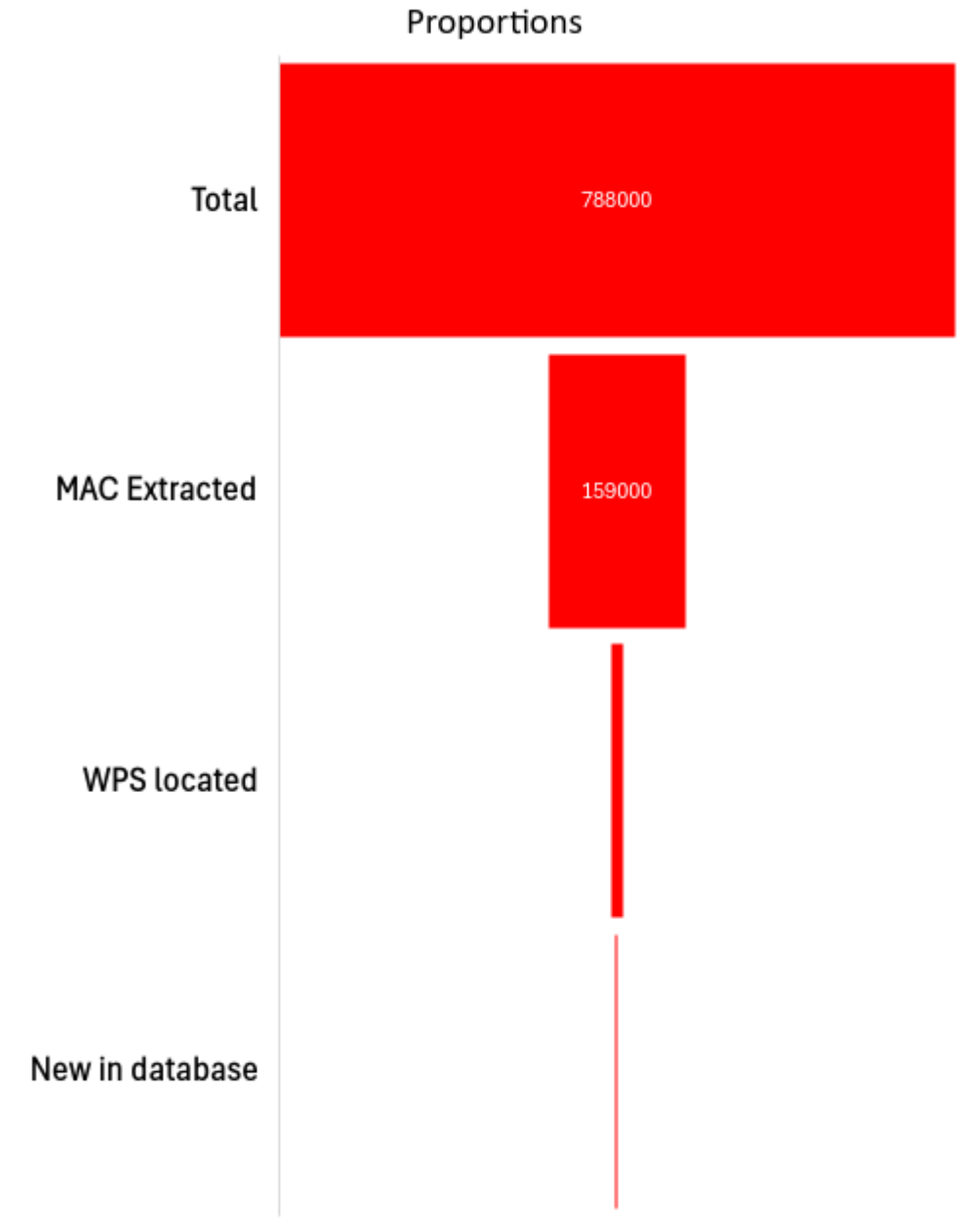


MAC Address Validation

SHORTCOMINGS

Out of 788,000 postings:

- MAC Extracted 20.18%
- WPS Located 1.65%
- New in database 0.165%



SHORTCOMINGS



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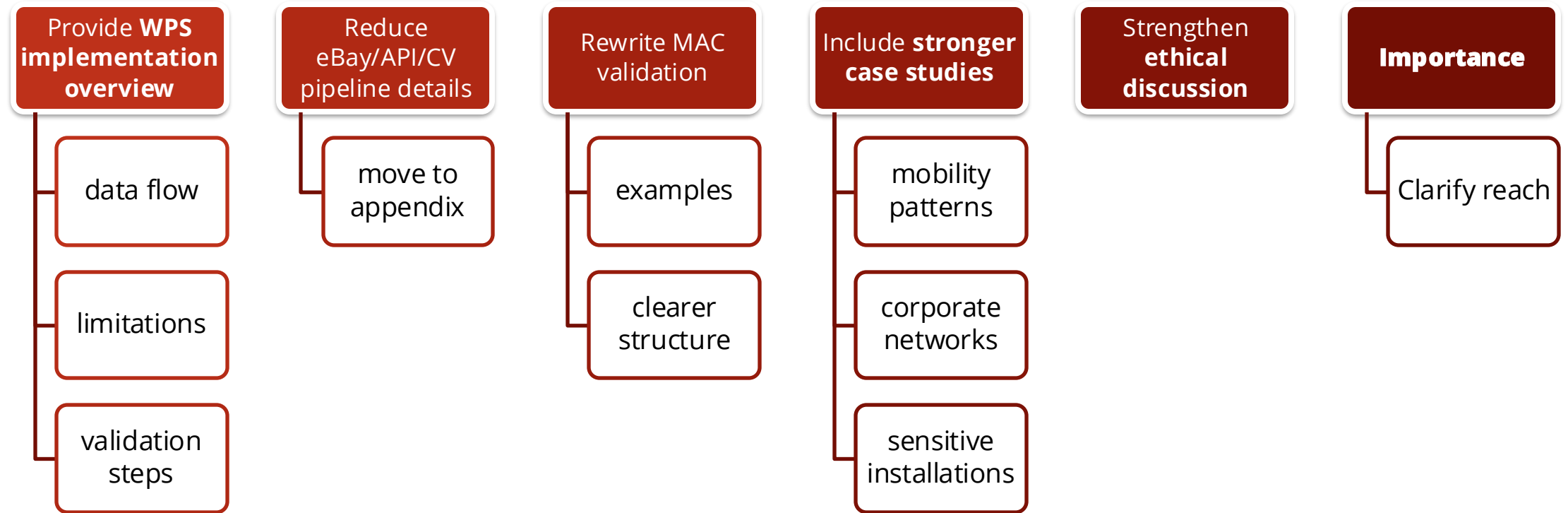


Relevance

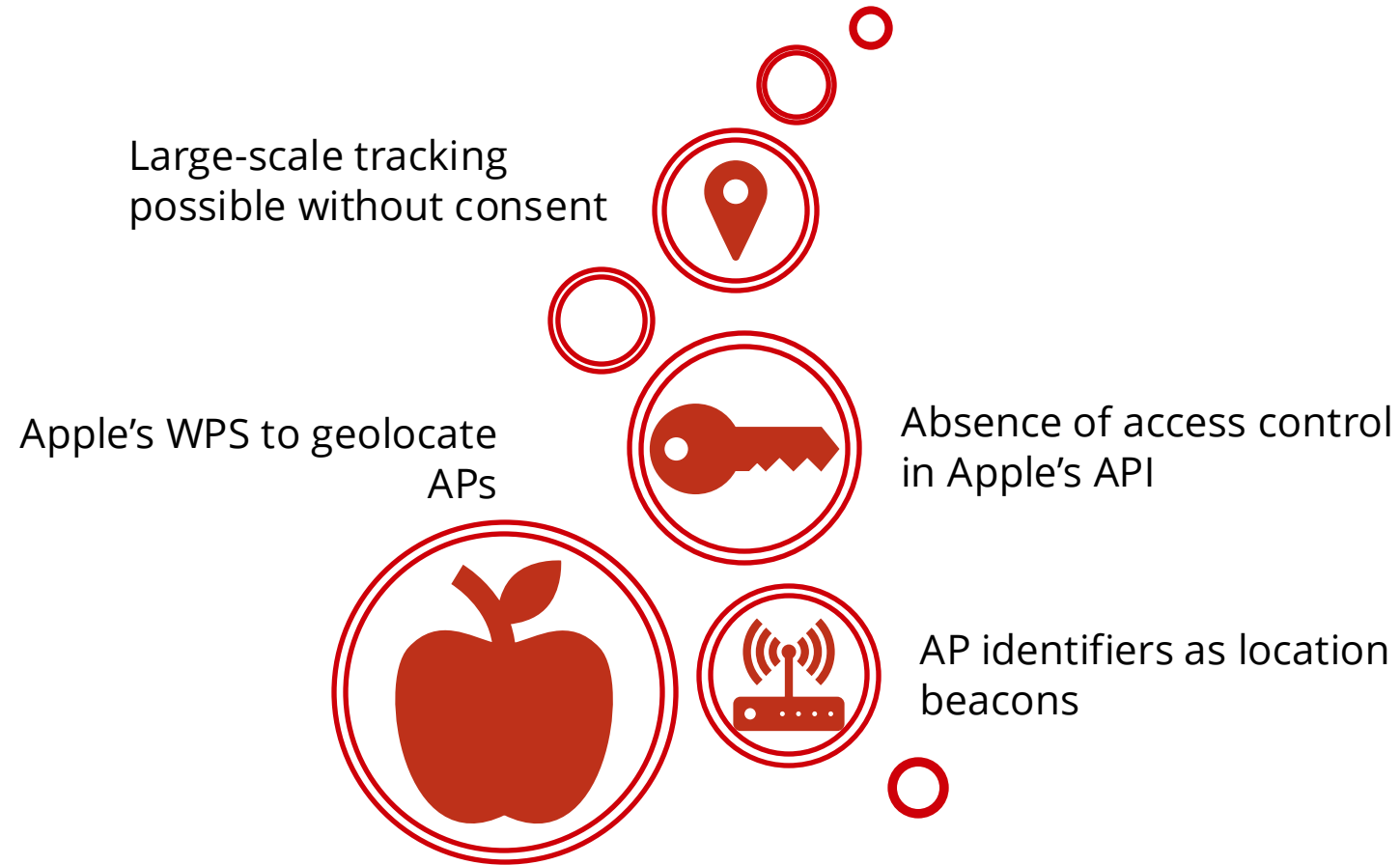


MAC Address Validation

SHORTCOMINGS



RELEVANT LECTURE: WEEK 7



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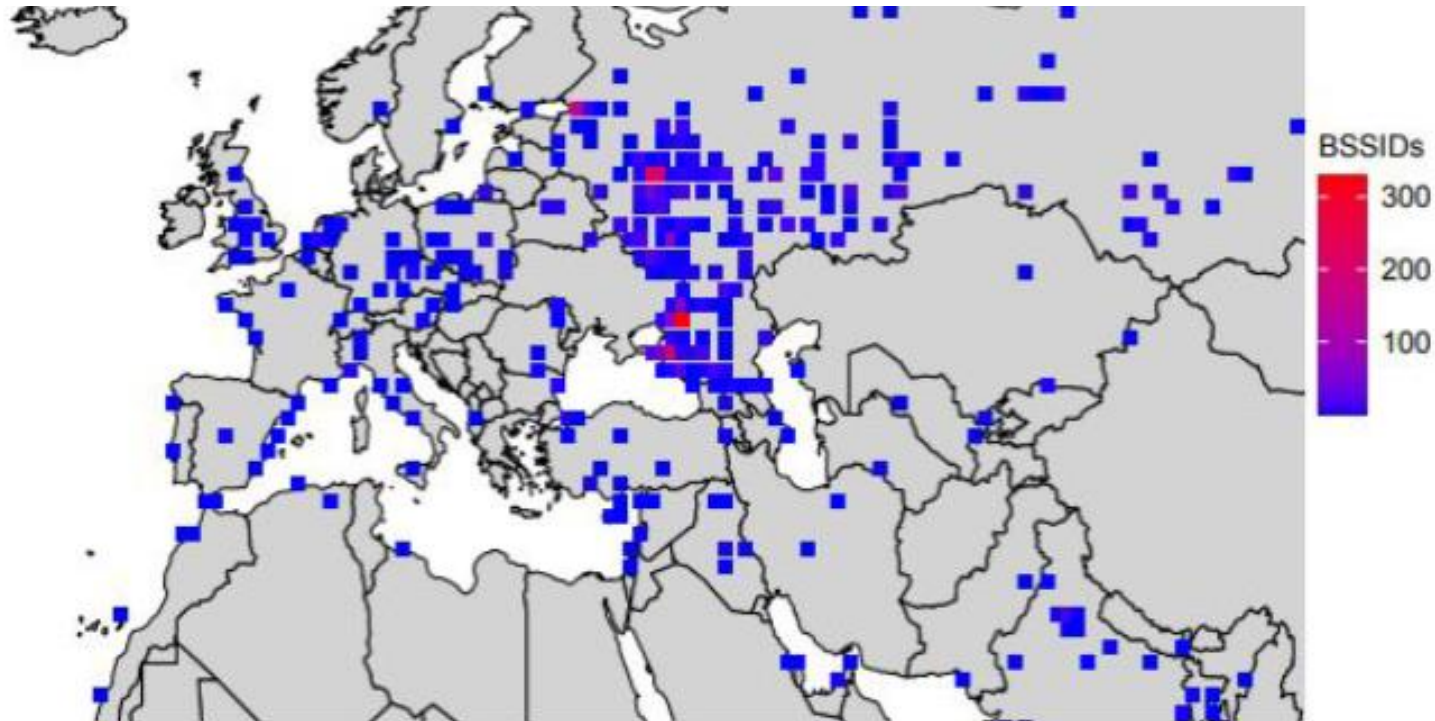
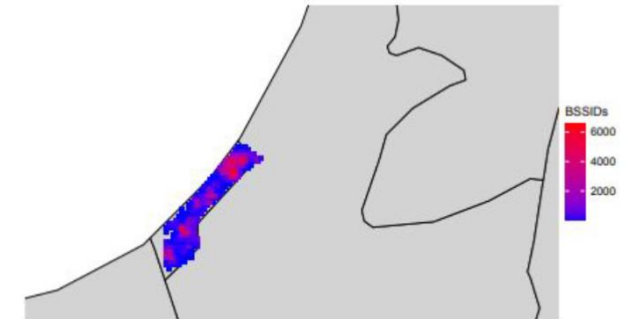
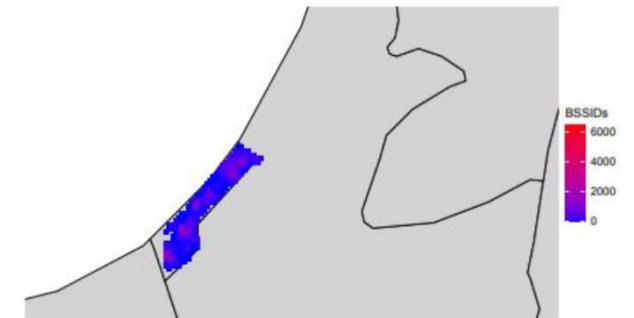


Figure 10: Heatmap of where BSSIDs that enter the Donbas and Crimea regions of Ukraine originate.



(a) Heat map of the BSSIDs in Gaza on 13 October 2023. Several hotspots are present in dense urban environments.



(b) Heat map of the BSSIDs in Gaza on 19 November 2023. Significantly lower density is seen in previous hotspots.

Figure 13: Heatmaps of BSSID density within the Gaza strip following the attack on Israel and Israel's response.

REFLECTION

Main takeaways

Public listings leak
mass amounts

Apple's WPS
capabilities

Sensitivity of
BSSIDs

New attack vector

Main issues

We believe they overstate
importance

Lack of ethical discussion

Not enough privacy focus